

Day 1 learning: Creating and Configuring a Virtual Machine (45 minutes)

Task 1: Create a Windows Virtual Machine

1. Log into Azure Portal (portal.azure.com)
2. Create a new Resource Group named "VM-Lab-RG"
3. Create a Windows Server 2022 VM with these specifications:
 - VM Name: WinServer-VM01
 - Region: Choose your nearest region
 - Size: Standard_B2s (2 vCPUs, 4 GB RAM)
 - Username: azureadmin
 - Password: (Create a strong password and note it down)
 - Public inbound ports: Allow RDP (3389)
 - OS disk type: Standard SSD
4. **Document:** Take screenshots of the VM creation process, especially the "Review + create" page showing all configurations.

Task 2: Explore VM Status and Properties

1. Once the VM is deployed, navigate to the VM overview page
2. Observe and document the following:
 - Current status (Running/Stopped/Deallocated)
 - Public IP address
 - Private IP address
 - DNS name
 - Resource group location
 - VM size and pricing
3. **Action:** Stop the VM and observe the status change. Note the difference between "Stopped" and "Stopped (deallocated)". Start it again.

Day 2: IP Address Configuration (30 minutes)

Task 1: Static IP Assignment

1. Navigate to your VM's networking settings
2. Change the private IP address from Dynamic to Static
3. Document the current private IP before and after the change

Task 2: Public IP Configuration

1. Go to the Public IP address resource associated with your VM
2. Change the assignment from Dynamic to Static
3. Add a DNS name label (e.g., yourname-winsrvr)
4. Test RDP connection using both IP address and DNS name

Questions to Answer:

- What happens to the public IP when you deallocate the VM with dynamic vs. static assignment?
- Why would you use a static private IP for a VM?

Day 3: Virtual Machine Disk Management (45 minutes)

Task 1: Add a Data Disk

1. While the VM is running, add a new data disk:
 - Name: DataDisk01
 - Size: 128 GB
 - Type: Standard SSD
 - Host caching: Read/write
2. Connect to the VM via RDP
3. Initialize and format the new disk in Windows:
 - Open Disk Management
 - Initialize the disk (GPT)
 - Create a new volume and assign drive letter E:
 - Format as NTFS
4. Create a test folder and file on the new drive

Task 2: Disk Performance Tiers

1. Navigate to the OS disk settings
2. Document the current performance tier and IOPS limits
3. **Question:** What's the difference between Standard HDD, Standard SSD, and Premium SSD?

Task 3: Disk Snapshots

1. Create a snapshot of your OS disk
2. Name it: OSDisk-Snapshot-[Today's Date]
3. Document the snapshot creation process

Day 4: Virtual Machine Management (40 minutes)

Task 1: VM Power States

Perform the following operations and document the time taken and billing impact:

1. **Stop** the VM (from Azure Portal)
2. Check the status
3. **Start** the VM
4. **Restart** the VM
5. **Stop (deallocate)** the VM
6. Check the status and note the difference

Task 2: Resize the Virtual Machine

1. While the VM is stopped (deallocated), resize it to Standard_B1s
2. Start the VM and verify it's running
3. **Question:** Can you resize a running VM? Try it and document what happens.

Task 3: VM Diagnostics and Monitoring

1. Enable Boot diagnostics on your VM
2. Navigate to Metrics and create a chart showing:
 - CPU percentage (last 1 hour)
 - Network in/out
3. Take a screenshot of the metrics

Task 4: Connect Using Multiple Methods

1. Connect via RDP using the public IP
2. Download the RDP file from Azure Portal and connect
3. Use Azure Bastion (if available) or document how it would work

Day 5: Cleanup and Cost Management (15 minutes)

Task 1: Review Costs

1. Navigate to Cost Management in your subscription
2. Filter costs by your resource group
3. Document the estimated daily cost of running your VM

Task 2: Proper Cleanup

1. Delete the snapshot you created
2. Delete the VM (select option to delete all associated resources)
3. Verify the resource group is empty

4. Delete the resource group

Deliverables

Submit a document (PDF or Word) containing:

1. **Screenshots** of:
 - VM creation summary page
 - VM overview showing status and IP addresses
 - Disk Management showing the attached data disk
 - Metrics dashboard
 - Snapshot creation confirmation
2. **Written answers** to all questions posed throughout the assignment
3. **Summary report** (300-500 words) covering:
 - What you learned about VM states and their billing implications
 - When to use static vs. dynamic IPs
 - Disk types and their use cases
 - Best practices for VM management
4. **Cost analysis**: Estimate the monthly cost of running a Standard_B2s VM 24/7 with 1 OS disk (128GB Standard SSD) and 1 data disk (128GB Standard SSD)